

2019 Ph H2 Q2

Section: Our Dynamic Universe

Topic: Forces — Abseiling equilibrium

A student of mass 55 kg abseils down a vertical wall using a rope. The rope is attached at the top and makes an angle of 15° to the building. (a) Calculate the student's weight. (b) Find the tension in the rope. (c) State and justify what happens to the tension if the rope angle decreases.

Worked solution

(a) The weight of the student:

$$W = m g = 55 \times 9.8 = 539 \text{ N.}$$

Answer: 539 N

(b) Vertical equilibrium: $T \sin(15^\circ) = W$.

$$T = W / \sin(15^\circ) = 539 / \sin(15^\circ) = 2083 \text{ N.}$$

Answer: 2080 N

(c) If the angle decreases, $\sin\theta$ becomes smaller. Since $T = W / \sin\theta$, the tension T increases.

Answer: The tension increases.

Final answers

(a) $W = 539 \text{ N}$

(b) $T = 2080 \text{ N}$

(c) Tension increases as rope angle decreases

Revision tips

- Weight $W = mg$, always acts vertically downwards.
- In equilibrium, resolve forces along vertical direction:
 $T \sin\theta = W$.
- Tension $= W / \sin\theta$.
- As angle decreases, $\sin\theta$ decreases, so T increases.
- Check limiting cases: if rope nearly vertical ($\theta \rightarrow 0$), tension $\rightarrow$ very large.